NARAYANA IIT-JEE ACADEMY-INDIA

JR STAR CO SC(MODEL-A&B) **Date: 25.07.2024**
Duration: 1 Hr **Max. Marks: 200**

CHEMISTRY NCERT TEST – 4

General Instructions:

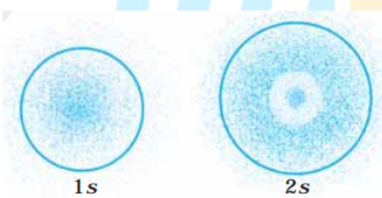
- **Attempt all the questions, each correct answer carries +4 and wrong answer -1.**

SYLLABUS: ATOMIC STRUCTURE

1. Which of the following conclusions could not be derived from Rutherford's α -particle scattering experiment?
   1) Most of the space in the atom is empty
   2) The radius of the atom is about 10^{-10} m while that of nucleus is 10^{-15} m
   3) Electrons move in a circular path of fixed energy called orbits
   4) Electrons and the nucleus are held together by electrostatic forces of attraction

2. Which of the following options does not represent ground state electronic configuration of an atom?
   1) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^8\ 4s^2$
   2) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^9\ 4s^2$
   3) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$
   4) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$

3. The probability density plots of 1s and 2s orbitals are given in Fig.



   The density of dots in a region represents the probability density of finding electrons in the region.
   On the basis of above diagram which of the following statements is incorrect?
   1) 1s and 2s orbitals are spherical in shape
   2) The probability of finding the electron is maximum near the nucleus.
   3) The probability of finding the electron at a given distance is equal in all directions
   4) The probability density of electrons for 2s orbital decreases uniformly as distance from the nucleus increases.

4. Which of the following properties of atom could be explained correctly by Thomson Model of atom?
   1) Overall neutrality of atom
   2) Spectra of hydrogen atom
   3) Position of electrons, protons and neutrons in atom.

4) Stability of atom

5. Which of the following is responsible to rule out the existence of definite paths or trajectories of electrons?

1) Pauli's exclusion principle.
2) Heisenberg's uncertainty principle
3) Hund's rule of maximum multiplicity
4) Aufbau principle

6. The pair of ions having same electronic configuration is __________.

1) Cr^{3+}, Fe^{3+}
2) Fe^{3+}, Mn^{2+}
3) Fe^{3+}, Co^{3+}
4) Sc^{3+}, Cr^{3+}

7. For the electrons of oxygen atom, which of the following statements is correct?

1) Z_{eff} for an electron in a 2s orbital is the same as Z_{eff} for an electron in a 2p orbital
2) An electron in the 2s orbital has the same energy as an electron in the 2p orbital
3) Z_{eff} for an electron in 1s orbital is the same as Z_{eff} for an electron in a 2s orbital.
4) The two electrons present in the 2s orbital have same spin quantum numbers m_s but of opposite sign

8. If travelling at same speeds, which of the following matter waves have the shortest wavelength?

1) Electron
2) Alpha particle (He^{2+})
3) Neutron
4) Proton

9. Out of the following pairs of electrons, identify the pairs of electrons present in degenerate orbitals :

(i) (a) $n=3,\ l=2,\ m_1=-2,\ m_s=-\frac{1}{2}$

(b) $n=3,\ l=2,\ m_1=-1,\ m_s=-\frac{1}{2}$

(ii) $n=3,\ l=1,\ m_1=1,\ m_s=+\frac{1}{2}$

(b) $n=3,\ l=2,\ m_1=1,\ m_s=+\frac{1}{2}$

(iii) (a) $n=4,\ l=1,\ m_1=1,\ m_s=+\frac{1}{2}$

(b) $n=3,\ l=2,\ m_1=1,\ m_s=+\frac{1}{2}$

(iv) (a) $n=3,\ l=2,\ m_1=+2,\ m_s=-\frac{1}{2}$

(b) $n=3,\ l=2,\ m_1=+2,\ m_s=+\frac{1}{2}$

1) (i) & (iv) only
2) (ii) & (iii) only
3) (iii) & (iv) only
4) (i), (ii), (iii) & (iv)

10. Which of the following statements concerning the quantum numbers are correct?

(i) Angular quantum number determines the three dimensional shape of the orbital.

(ii) The principal quantum number determines the orientation and energy of the orbital.

(iii) Magnetic quantum number determines the size of the orbital.

(iv) Spin quantum number of an electron determines the orientation of the spin of electron relative to the chosen axis.

1) (i) & (ii) only   2) (i) & (iv) only   3) (ii) & (iii) only   4) (i), (ii), (iii) & (iv)

11. Match the following species with their corresponding ground state electronic configuration.

Atom / Ion	**Electronic configuration**
(i) Cu	(a) $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^{10}$
(ii) Cu^{2+}	(b) $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^{10}\,4s^2$
(iii) Zn^{2+}	(c) $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^{10}\,4s^1$
(iv) Cr^{3+}	(d) $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^9$
	(e) $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^3$

1) (i)-(c), (ii)-(d), (iii)-(a), (iv)-(e)   2) (i)-(d), (ii)-(c), (iii)-(b), (iv)-(a)

3) (i)-(a), (ii)-(b), (iii)-(d), (iv)-(e)   4) (i)-(c), (ii)-(a), (iii)-(b), (iv)-(b)

12. Match the following

Column I	**Column II**
(i) X – rays	(a) $v = 10^0 - 10^4\,Hz$
(ii) UV	(b) $v = 10^{10}\,Hz$
(iii) Long radio waves	(c) $v = 10^{16}\,Hz$
(iv) Microwave	(d) $v = 10^{18}\,Hz$

1) (i)-(a), (ii)-(b), (iii)-(c), (iv)-(d)   2) (i)-(d), (ii)-(c), (iii)-(a), (iv)-(b)

3) (i)-(d), (ii)-(c), (iii)-(b), (iv)-(a)   4) (i)-(c), (ii)-(d), (iii)-(a), (iv)-(b)

13. Assertion (A) : All isotopes of a given element show the same type of chemical behaviour.

Reason (R) : The chemical properties of an atom are controlled by the number of electrons in the atom

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true but R is not the correct explanation of A.

3) A is true but R is false.

4) Both A and R are false.

14. Assertion (A) : Black body is an ideal body that emits and absorbs radiations of all frequencies.

Reason (R) : The frequency of radiation emitted by a body goes from a lower frequency to higher frequency with an increase in temperature.

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true but R is not the correct explanation of A.

3) A is true but R is false.

4) Both A and R are false.

15. Assertion (A) : It is impossible to determine the exact position and exact momentum of an electron simultaneously.

Reason (R) : The path of an electron in an atom is clearly defined.

1) Both A and R are true and R is the correct explanation of A

2) Both A and R are true but R is not the correct explanation of A.

3) A is true but R is false.

4) Both A and R are false.

16. The extra stability of half-filled and completely filled subshell is due to

(A) relatively small shielding (B) smaller coulombic repulsion energy

(C) larger exchange energy

The correct option is:

1) A & B only 2) A & C only 3) B & C only 4) A, B & C

17. Values of work function (W_0) for a few metals are given below:

Metal	Li	Na	K	Mg	Cu	Ag
W_0 / eV	2.42	2.3	2.25	3.7	4.8	4.3

The number of metals which will show photoelectric effect when light of wavelength 400 nm falls on it is ___

Given : $h = 6.6\times10^{-34}\,Js$

$c = 3\times10^{8}\,ms^{-1}$

$e = 1.6\times10^{-19}\,C$

1) 5 2) 4 3) 6 4) 3

18. Assertion (A): In the photoelectric effect, the electrons are ejected from the metal surface as soon as the beam of light of frequency greater than threshold frequency strikes the surface.

Reason (R): When the photon of any energy strikes an electron in the atom, transfer of energy from the photon to the electron takes place.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Both A and R are correct but R is not the correct explanation of A

2) A is correct but R is not correct

3) Both A and R are correct but R is the correct explanation of A

4) A is not correct but R is correct

19. The energy of one mole of photons of radiation of wavelength 300 nm is

(Given: $h = 6.63\times10^{-34}\,Js, N_A = 6.02\times10^{23}\,mol^{-1}$, $c = 3\times10^{8}\,ms^{-1}$)

1) 235 $kJ\,mol^{-1}$ 2) 325 $kJ\,mol^{-1}$ 3) 399 $kJ\,mol^{-1}$ 4) 435 $kJ\,mol^{-1}$

20. A 50 watt bulb emits monochromatic red light of wavelength of 795 nm. The number of photons emitted per second by the bulb is

(Given: $h = 6.63\times10^{-34}\,Js\, and\, c = 3.0\times10^{8}\,ms^{-1}$)

1) 5×10^{20} 2) 2×10^{20} 3) 2×10^{18} 4) 3×10^{20}

21. What is the work function of the metal if the light of wavelength 4000 A^o generates photoelectrons of velocity $6\times10^5\ ms^{-1}$ from it?

(Mass of electron = $9\times10^{-31}\ kg$, Velocity of light = $3\times10^8\ ms^{-1}$, Planck's constant = $6.626\times10^{-34}\ Js$, Charge of electron = $1.6\times10^{-19}\ JeV^{-1}$)

1) 0.9 eV
2) 3.1 eV
3) 2.1 eV
4) 4.0 eV

22. The energy of an electron in the first Bohr orbit of hydrogen atom is $-2.18\times10^{-18}\ J$. Its energy in the third Bohr orbit is ____

1) $\frac{1}{27}$ of this value
2) one third of this value
3) three times of this values
4) $\frac{1}{9}th$ of this value

23. The shortest wavelength of hydrogen atom in Lyman series is λ. The longest wavelength in Balmer series of He^+ is

1) $\frac{5}{9\lambda}$
2) $\frac{9\lambda}{5}$
3) $\frac{36\lambda}{5}$
4) $\frac{5\lambda}{9}$

24. Which one of the following about an electron occupying the 1s orbital in a hydrogen atom is incorrect? (The Bohr radius is represented by a_0).

1) The probability density of finding the electron is maximum at the nucleus
2) The electron can be found at a distance 2 a_0 from the nucleus
3) The magnitude of the potential energy is double that of its kinetic energy on an average
4) The total energy of the electron is maximum when it is at a distance a_0 from the nucleus

25. The limiting line in Balmer series will have a frequency of (Rydberg constant, 3.29×10^{15} cycle/s)

1) $8.22\times10^{14}\ s^{-1}$
2) $3.29\times10^{15}\ s^{-1}$
3) $3.65\times10^{14}\ s^{-1}$
4) $5.26\times10^{13}\ s^{-1}$

26. If the radius of the first orbit of hydrogen atom a_0, then de-Broglie's wavelength of electron in 3rd orbit is

1) $\frac{\pi a_0}{6}$
2) $\frac{\pi a_0}{3}$
3) $6\pi a_0$
4) $3\pi a_0$

27. An accelerated electron has speed of $5\times10^6\ ms^{-1}$ with an uncertainty of 0.02%. The uncertainty in finding its location while in motion is $x\times10^{-9}\ m$. The value of x is ____ (Nearest integer) [use mass of electron = $9.1\times10^{-31}\ kg, h = 6.63\times10^{-34}\ Js, \pi = 3.14$]

1) 58
2) 68
3) 6
4) 44

28. Compare the energies of following sets of quantum numbers for multielecton system

(A) n = 4, l = 1
(B) n = 4, l = 2
(C) n = 3, l = 1
(D) n = 3, l = 2
(E) n = 4, l = 0

Choose the correct answer from the options given below:

1) (B) > (A) > (C) > (E) > (D)
2) (E) > (C) > (D) < (A) < (B)

3) (E) > (C) > (A) > (D) > (B) 4) (C) < (E) < (D) < (A) < (B)

29. The maximum number of orbitals which can be identified with n = 4 and $m_l = 0$ is ______

1) 6 2) 4 3) 2 4) 1

30. The number of nodal surface/s and angular node/s for 3p orbital is

1) 3 & 2 2) 2 & 1 3) 1 & 1 4) 4 & 1

31. The four quantum numbers for the electron in the outer most orbital of potassium (atomic no. 19) are

1) n = 4, l = 2, m = -1, s = $+\frac{1}{2}$ 2) n = 4, l = 0, m = 0, s = $+\frac{1}{2}$

3) n = 3, l = 0, m = 1, s = $+\frac{1}{2}$ 4) n = 2, l = 0, m = 0, s = $+\frac{1}{2}$

32. Statement I: The orbitals having same energy are called as degenerate orbitals.

Statement II: In hydrogen atom, 3p and 3d orbitals are not degenerate orbitals.

In the light of the above statements, choose the most appropriate answer from the options given below:

1) Statement I is true but statement II is false

2) Both statement I and statement II are true

3) Both statement I and statement II are false

4) Statement I is false but statement II is true

33. The number of given orbitals which have electron density along the axis is ____

$p_x, p_y, p_z, d_{xy}, d_{yz}, d_{z^2}, d_{x^2-y^2}$

1) 3 2) 2 3) 5 4) 4

34. Given

(A) n = 5, $m_l = +1$

(B) n = 2, $l = 1, m_l = -1, m_s = -1/2$

The maximum number of electron(s) in an atom that can have the quantum numbers are given in (A) and (B) are respectively:

1) 25 and 1 2) 8 and 1 3) 2 and 4 4) 4 and 1

35. The number of d-electrons retained in Fe^{2+} (At.no. of Fe = 26) ion is

1) 4 2) 5 3) 6 4) 3

36. Calculate the wavelength of a light wave whose period is $2.0\times10^{-10}s$

1) $6\times10^{-2}m$ 2) $3\times10^{-2}m$ 3) $1.2\times10^{-1}m$ 4) $9\times10^{-2}m$

37. A 25 watt bulb emits monochromatic yellow light of wavelength of 0.57 μm. Calculate the rate of emission of quanta per second.

1) $7.1\times10^{19}s^{-1}$ 2) $34.8\times10^{20}s^{-1}$ 3) $25\times10^{19}s^{-1}$ 4) $4\times10^{19}s^{-1}$

38. The mass of an electron is $9.1\times10^{-31}kg$. If its K.E. is $3.0\times10^{-25}J$, what is the wavelength of that electron?

1) $9\times10^{-7}m$ 2) $8\times10^{-6}m$ 3) $9\times10^{-6}m$ 4) $5\times10^{-7}m$

39. An element 'X' with mass number 81 contains 31.7% more neutrons as compared to protons. Assign the atomic symbol.

1) $^{81}_{35}X$ 2) $^{35}_{81}X$ 3) $^{81}_{40}X$ 4) $^{40}_{81}X$

40. For an electronic transition occurring in hydrogen atom, the energy absorbed is maximum in the case of

1) $2 \rightarrow 1$ 2) $\infty \rightarrow 1$ 3) $5 \rightarrow 2$ 4) $1 \rightarrow 2$

41. Radius of certain orbit in hydrogen atom is $4.76\,A^o$, total energy of electron revolving in that orbit will be

1) $-2.42\times10^{-19}\,J$ 2) $-7.29\times10^{-19}\,J$ 3) $-6.54\times10^{-18}\,J$ 4) $-1.36\times10^{-19}\,J$

42. Mass ratio of two particles x and y is 2:1 and the ratio of their kinetic energies is 2:9. de Broglie wavelength ratio of x and y will be

1) 4 : 3   2) 9 : 4   3) 3 : 2   4) 2 : 3

43. Charge present on 1 mole of Na^+ ion is

1) 96500 C   2) $3\times96500\,C$ 3) $4.8\times10^{-19}\,J$ 4) 965 C

44. Which one of the following statement is incorrect?

1) Electromagnetic wave theory cannot explain photoelectric effect

2) Electromagnetic wave theory cannot explain heat capacity of solids as a function temperature

3) Emission spectrum can be continuous can discontinuous

4) Absorption spectrum can be continuous (or) discontinuous.

45. Ratio of wavelengths of first line of Lymen series of H-spectrum to second line of Balmer series of H-spectrum is

1) 1 : 2   2) 2 : 1   3) 1 : 4   4) 4 : 1

46. According to Rutherford"s theory

Assertion (A): A few α - particles were deflected when passed through gold foil.

Reason(R): Positively charged α - particles are repelled by nucleus of an atom.

1) Both (A) and (R) are true and (R) is the correct explanation of (A)

2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

3) (A) is true but (R) is false

4) Both (A) and (R) are false

47. Radius of first orbit of H-atom is 'x'. Circumference of 4th orbit in H-atom would be

1) 32 πx 2) 64 πx 3) 16 πx 4) 128 πx

48. Identify correct statements from the following

(A) Intensity of black body radiation varies with temperature

(B) Maximum intensity of black body radiation shifts to lower frequency at higher temperature

(C) Emission of black body radiations is discontinuous

1) A, B, C   2) A, C only   3) A, B only   4) B, C only

49. Orbital angular momentum of 3d electron is

1) 0 2) $\sqrt{2}\,\hbar$ 3) $\sqrt{6}\,\hbar$ 4) $\sqrt{12}\,\hbar$

50. If the uncertainty in the position is two times that of momentum for a moving particle then the uncertainty in the velocity of the particle will be

1) $\frac{h}{4\pi}$ 2) $\sqrt{\frac{h}{8\pi m^2}}$ 3) $\sqrt{\frac{h}{4\pi m}}$ 4) $\frac{h}{2\pi m}$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AIR 76
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AIR 82
AIR 85
AIR 87
AIR 89
AIR 90
AIR 91
AIR 94
AIR 99
HIGHLIGHTS
Below 20 All India Open Category Ranks 7 Ranks
Below 100 All India Open Category Ranks 31 Ranks
Below 500 All India Open Category Ranks 96 Ranks
Below 1000 All India Open Category Ranks 179 Ranks
Below 100 All India Category Ranks 127 Ranks
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28
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34
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40
VISHARAD SRIVASTAVA
47
AMOGH AGRAWAL
62
JAGANNADHAM MOHITH
64
SRI SURYA VARMA DATLA
66
YERUVA KETAN REDDY
69
MAJJI RISHI VARDHAN
78
SAMBARI MANIDEEP KUMAR
82
DEVANSH GATTANI
84
PATHRI VIDYA PRAVEEN
85
ABHINAV CHHABRA
88
P REHAN
90
AKSHAT KHANDELWAL
Many More...
6
in Top 10
Open Category
7
in Top 12
Open Category
28
in Top 100
Open Category
891
in Top 1000
All Category